MATH317 · NUMERICAL METHODS

Chapter 1.1 Preliminary Remarks

Midterm summary · Lecture 2

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What's inside

01 What is this course about?

02 Significant digits

03 Absolute error and Relative error

04 Accuracy and Precision

05 Rounding and Chopping

06 Horner's algorithm

07 Deflation

08 How it shows up in the midterm

09 Common mistakes

10 Check yourself

01 What is this course about?

A calculator does not know the answer in advance. It builds an answer that is very close to the true one using simple operations.

This course teaches two things: how to approximate, and how large the error of the approximation is.

The big idea

Replace a hard problem with a sequence of easier ones. Every numerical method works this way.

02 Significant digits

Significant digits are the digits that carry meaning. Start counting from the first nonzero digit on the left.

Number	Significant digits	Why
`0.0035`	2	Leading zeros do not count.
`30.16`	4	A zero between digits counts.
`12.345`	5	Every digit counts.

Slide example: why the number of digits matters

Solve the same system three times, keeping a different number of digits each time:

```
0.1036 x + 0.2122 y = 0.7381
```

```
0.2081 x + 0.4247 y = 0.9327
```

Digits kept	Value of y
3	−547
4	343.9
10	356.29

The lesson

Do not round in the middle of a calculation. Carry full precision until the last step.

03 Absolute error and Relative error

Let the exact value be α and the approximate value be β.

$$\text{Absolute error} = |\alpha - \beta|$$

$$\text{Relative error} = \frac{|\alpha - \beta|}{|\alpha|}$$

For a percentage, multiply the relative error by 100.

Slide example

1 $x = 0.00347$ **rounded to** 0.0035

$$\text{Absolute error} = |0.00347 - 0.0035| = 0.00003 = 0.3 \times 10^{-4}$$

$$\text{Relative error} = \frac{0.00003}{0.00347} = 0.865 \times 10^{-2}$$

2 `y = 30.158` **rounded to** `30.16`

$$\text{Absolute error} = |30.158 - 30.16| = 0.002 = 0.2 \times 10^{-2}$$

$$\text{Relative error} = \frac{0.002}{30.158} = 0.66 \times 10^{-4}$$

Notice

The absolute error is larger in the second case, yet its relative error is much smaller.

The relative error is the better measure of accuracy.

Why measure the error?

- To judge how accurate a result is.
- To decide when an iterative method should stop (**Stopping criteria**).

04 Accuracy and Precision

Term	Meaning
Accurate to n decimal places	You can trust n digits to the right of the decimal point.
Accurate to n significant digits	You can trust n digits counted from the first nonzero digit.

Example: `12.345` m is accurate to 3 decimal places and has 5 significant digits.

Accuracy in arithmetic

A result is never more accurate than the least accurate number used.

```
1.2 + 3.45 = 4.6          (2 significant digits)
```

```
1.23 × 4.5 = 5.535        (really only 2 significant digits)
```

05 Rounding and Chopping

Method	What it does
Chopping	Drop every digit after the n-th one and leave the rest unchanged.
Rounding	Replace the number by the nearest n-digit number, so the error is as small as possible.

Slide examples (to two digits)

Number	Rounding	Chopping
0.217	0.22	0.21
0.365	0.37	0.36
0.475	0.48	0.47
0.592	0.59	0.59

Rounding is usually preferred because its error is smaller.

Appeared in the midterm

A True/False statement in midterm 472 described the difference between rounding and chopping. The answer was True.

06 Horner's algorithm

A fast way to evaluate a **Polynomial** at a point without computing any powers. It is also called **Nested multiplication** or **Synthetic division**.

$$p(x) = 5 + 3x - 7x^2 + 2x^3 = 5 + x(3 + x(-7 + x(2)))$$

The table method

1 Write the coefficients from the highest power down.

If a power is missing, write `0` in its place.

2 Bring the first coefficient down unchanged.

3 Multiply by `r`, write the result under the next coefficient, and add.

4 Repeat to the last column. The final number is `p(r)`.

Worked example

Find `P(3)` for $P(x) = 4x^3 - 8x^2 + 14x - 32$.

	4	−8	14	−32
3 |		12	12	78
	4	4	26	46

```
P(3) = 46
```

How each number is found: `4×3 = 12` then `−8+12 = 4`. Next `4×3 = 12` then `14+12 = 26`. Next `26×3 = 78` then `−32+78 = 46`.

Watch out

In the midterms the polynomial always has a missing power. Example: `3x³ − 4x² − 12` has coefficients `3, −4, 0, −12`.

07 Deflation

If the last number in the Horner table is zero, then `r` is a **Root** of the polynomial.

The other numbers in the last row are then the coefficients of a new polynomial of one degree less. This process is called **Deflation**.

```
p(x) = (x − r) · q(x)
```

In words: the factor `(x − r)` is removed, and the remaining roots are the roots of `q(x)`.

08 How it shows up in the midterm

Question style	Midterms	Marks
Compute the absolute and relative errors	461 · 462 · 471	4–7
Evaluate `P(r)` with Horner	462 · 471 · 472	3
True / False	472	7

The error question from midterm 471 (7 marks)

Let x = 0.54617 and y = 0.54601. Use four-digit arithmetic to approximate x − y and determine the absolute and relative errors using rounding.

Exact: $r = x - y = 0.00016$

Round x, y: $\tilde{x} = 0.5462 \quad \tilde{y} = 0.5460$

Approximate: $\tilde{r} = 0.5462 - 0.5460 = 0.0002$

Absolute error = $|0.00016 - 0.0002| = 0.00004$

$$\text{Relative error} = \frac{0.00004}{0.00016} = 0.25$$

The relative error is 25%, which is large. The cause is subtracting two nearly equal numbers.

09 Common mistakes

Mistake	Fix
Forgetting the zero for a missing power in Horner	Write every coefficient, including zeros.
Dividing the relative error by the approximate value	Divide by the exact value.
Rounding in the middle of the work	Round only at the end, unless the question says otherwise.
Mixing up rounding and chopping	**Chopping** drops digits. **Rounding** goes to the nearest value.

10 Check yourself

1. Exact value 3.14159, approximation 3.14. Find the absolute and relative errors.
2. Write 0.4768 to two digits, once by rounding and once by chopping.
3. Use Horner to find $p(3)$ for $p(x) = 2x^3 - 7x^2 + 3x + 5$.
4. Use Horner to find $p(-1)$ for $p(x) = 2x^4 - 5x^2 + x - 7$.

Solutions

1) Absolute $= |3.14159 - 3.14| = 0.00159$

$$\text{Relative} = \frac{0.00159}{3.14159} = 5.06 \times 10^{-4}$$

2) Rounding: 0.48 Chopping: 0.47

3)

	2	-7	3	5	
3 |		6	-3	0	
	2	-1	0	5	$p(3) = 5$

4)

	2	0	-5	1	-7	
-1 |		-2	2	3	-4	
	2	-2	-3	4	-11	$p(-1) = -11$